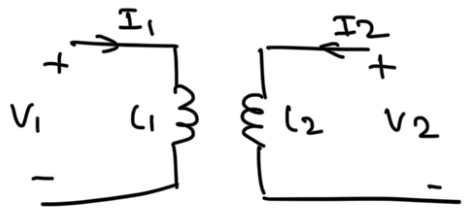
MI transformer:



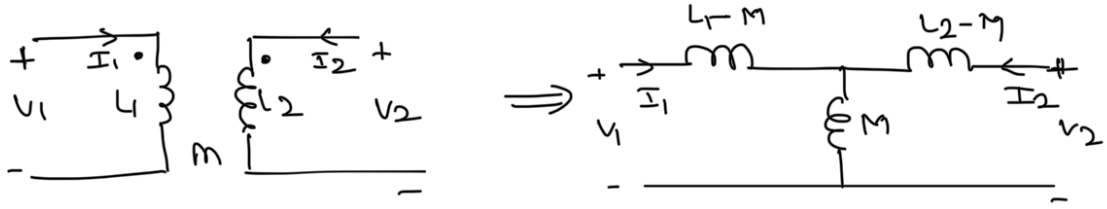
$$V_1 = L_1\frac{di_1}{dt} + M_{12}\frac{di_2}{dt}$$

$$V_2 = L_2\frac{di_2}{dt} + M_{21}\frac{di_1}{dt}$$

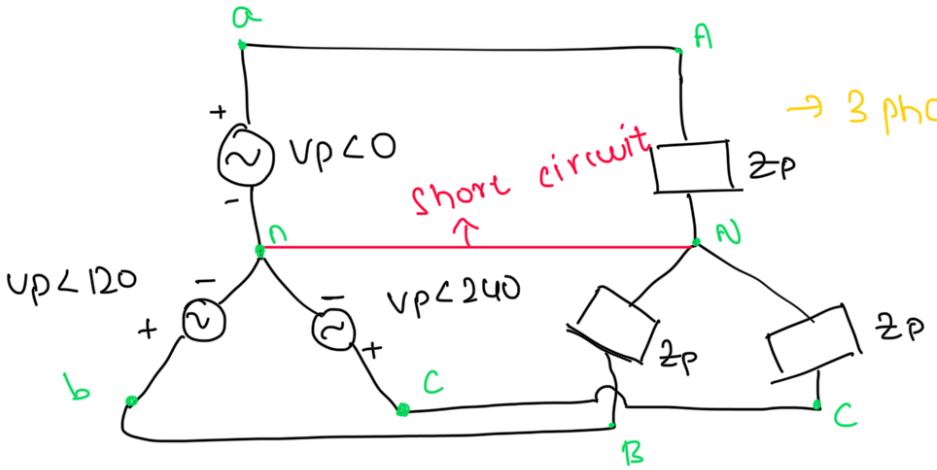
By symmetry of magnetic flux and energy flux

$$|M_{12}| = |M_{21}| = M$$

increasing current entering at • terminal of one inductor produces positive voltage at the • terminal of the other



Three phase: Ensure uniform supply of power



→ 3 phase Y source Y load

$$I_{aA} = \frac{V_p\angle 0}{Z_p}$$

$$I_{bB} = \frac{V_p\angle 120}{Z_p}$$

$$I_{cC} = \frac{V_p\angle 240}{Z_p}$$

$$I_{nN} = 0$$

$$I_{aA} + I_{bB} + I_{cC} = V_p/Z_p$$